Primitive Axiomatics of Finite Cohomological Obstruction Theory

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Abstract

This paper states the foundation for the obstruction calculus and separates proved structure from observer-dependent assumptions. Starting from scalar observations, localized comparison is modeled by an evaluation relation whose reflexive-transitive closure is an **Evaluation Preorder**. Pairwise inconsistency over a chosen simplicial carrier gives a cochain complex. Once an observer has selected an edge-decomposable, replica-extensive aggregation law and compatible local gauges, the resulting scalar obstruction is the quotient norm of the observed defect field modulo exact repair. In finite dimensions this quotient admits a bounded divergence-free dual witness. The paper replaces "Universal Forcing" with **Conditional Rigidity**: no metric geometry is claimed without specifying the observer assumptions that select it.

Position: Foundational. All subsequent papers (000–General Theory) inherit from this document.

1 §1.0 — Carrier-General Defect Systems

1.1 Definition

The primitive scalar base model defines the local defect between two scalar continuous observations functionally as $D(x, y) := y - x$. However, the continuous operator obstruction engine inherently encompasses a substantially wider analytical generalization.

A generic evaluative network structurally requires strictly a **Carrier-General Defect System**:

- A carrier space K formally assigning exact local state spaces $\mathcal{F}(v)$.
- Evaluative mapping comparisons $r_{b,a} : \mathcal{F}(b) \to \mathcal{F}(a)$.
- A strictly isolatable edge defect physical obstruction $\delta_e(s) \in D_e$.
- A strictly faithful continuous local measurement gauge $j_e : D_e \to \mathbb{R}_{\geq 0}$.

The primitive scalar baseline is recovered when the following restrictions are imposed:

1. **Antisymmetry:** $D(x, y) = -D(y, x)$
2. **Linearity:** $D(x, y) + D(y, z) = D(x, z)$
3. **Faithfulness:** $D(x, y) = 0 \iff x = y$

4. **Strict Scalar limits:** physical observations evaluate strictly exclusively across scalar bounding boundaries ($\mathbb{R}$).

Correction

The theory is carrier-general at the level of defects, but strictly **scalar-valued** at the level of measurement ($j_e : D_e \to \mathbb{R}_{\geq 0}$). Without the scalar evaluation metric, the magnitude quotient metric Φ_1 is undefined.

Status: CONDITIONAL GENERALIZATION

The exact-sequence formalism extends to any compatible carrier with edge-local defects and scalar-valued gauges. The scalar $\mathbb{R}$ model is the base example, not a proof that all physical or computational carriers reduce to scalar differences.

2 §1.1 — The Trace Instrumentation Bridge (Lawvere Shadow)

Abstract categorical undecidability generating a discrete paradox (producing $\perp$) is disconnected from a continuous scalar defect. We do not claim a full "Ghost Functor" mapping abstract undecidability globally into metric coefficients. Instead, we establish a **Trace Instrumentation Bridge** mapping finitely accumulated partial evaluation histories strictly into directed metric obstruction boundaries.

1. **Finite Trace Extraction:** Let τ be a finitely recorded trace of partial evaluation attempts defined by evaluation states V, traversed dependency edges E, and an **externally defined** cost signal scoring success/failure: $\omega_\tau(x \to y) = \text{cost}(\text{eval}(y) \text{ rel eval}(x))$. Note carefully that the functional cost scoring structure sits as an injected boundary explicit relation outside the generated derived geometry.

2. **Trace Instrumentation Map:** The map $\mathcal{L}_{\text{trace}}(\tau) = (P_\tau, \omega_\tau)$ constructs the physical graph representing exactly what the evaluator verifiably attempted.

3. **Trace Lawvere Shadow Theorem:** Given a finite observed trace τ containing a closed dependency loop whose edge costs cannot be exactly resolved ($\omega_\tau \notin \text{im}(d_0)$), the induced obstruction carrier immediately guarantees strictly isolated positive defect $\Phi_1(P_\tau, \omega_\tau) > 0$.

The topological metric does not magically score "the truth value of a paradox". Rather, when a system generates an evaluative loop incapable of syntactic exact closure, evaluating the trace network natively measures the *structural trace inconsistency.*

Status: STRUCTURAL CONJECTURE / INSTRUMENTATION BRIDGE

The Lawvere bridge is not a derivation of Φ_1 natively from undecidability. It is a strict instrumentation principle: finite traces of partial self-evaluation can be encoded as directed obstruction carriers. When such traces record non-exact cyclic inconsistency, the quotient norm Φ_1 captures the irreducible residual. Thus Φ_1 measures the metric shadow of a recorded evaluation failure, not abstract undecidability itself.

3 §1.2 — Evaluation Preorder Emergence

3.1 Forcing Chain

Observation is local. Not all pairs of states are directly comparable. Modeling observed comparability as a relation and closing under identity and composable paths gives:

localization of observation
$\to$ dependency relation
$\to$ reflexivity + transitivity
$\to$ **Evaluation Preorder** (after reflexive-transitive closure)
$\to$ finite directed graph $P = (V, E)$

Correction on Poset Strictness: Directed local evaluations induce relations of comparable operational flow, creating an **evaluation preorder**, not a strict Poset. Acyclic propagation paths yield stratified posets, but cyclic observation structures map cleanly to irreducible directed graph topologies. These cyclic boundaries evaluate rigorously to positive residual $\Phi_1 > 0$ and represent the carriers of stable mathematical breakdown, rather than a contradiction of the graph grammar.

3.2 Simplicial Bridge

The preorder graph P induces a simplicial structure:

- 0-simplices = vertices V (observation sites)
- 1-simplices = edges E (pairwise comparisons)
- 2-simplices = triangles $S_2(P)$ (triple consistency tests)
- n-simplices = $S_n(P)$ ($n+1$-fold consistency)

This simplicial structure canonically produces the cochain spaces of §1.3:

- $C^0(P) := \mathbb{R}^V$ (local state assignments)
- $C^1(P) := \mathbb{R}^E$ (pairwise inconsistencies)
- $C^2(P) := \mathbb{R}^{S_2}$ (triple obstruction)

Status: MODEL-DERIVED

Local directed observation gives a relation. Passing to its reflexive-transitive closure gives a preorder; choosing a nerve/simplicial thickening then gives the cochain domains. Without that closure and thickening choice, a directed graph alone does not canonically supply all higher cochains.

4 §1.3 — Cohomological Split

4.1 Coboundary Operator

The graph $P = (V, E)$ induces the degree-0 coboundary:

$$d_0 : C^0(P) \to C^1(P), \quad (d_0 f)_e = f(b) - f(a) \text{ for } e = (a \to b) \tag{1}$$

This is exactly the primitive difference operator D lifted to the cochain level. When 2-simplices $S_2(P)$ are defined, the degree-1 coboundary $d_1 : C^1(P) \to C^2(P)$ satisfies:

$$d_1 \circ d_0 = 0 \tag{2}$$

This is not assumed — it is a consequence of antisymmetry of D and the simplicial structure.

4.2 Quotient Magnitude (The "Optimization" Reframe)

Given an observed inconsistency signal $\delta \in C^1(P)$, we define **Gauge Equivalence**: two fields are equivalent $[\delta] = [\delta']$ if they differ only by a trivial internal reference shift $\delta - \delta' \in \mathrm{im}(d_0)$.

The topological magnitude of this physical class is rigorously the distance to the exact subspace. In Banach spaces, the norm of a quotient class is analytically uniquely defined as the infimum distance:

$$\Phi_1 := \|[\delta]\|_{C^1/\,\mathrm{im}(d_0)} = \inf_{f \in C^0} \|\delta - d_0 f\|_J, \quad h_1 := \delta - d_0 f^* \tag{3}$$

Interpretation: There is no extra variational principle beyond the quotient norm. However, Φ_1 is not primitive: it is induced by the prior choice of observer gauge J. In finite-dimensional settings, one may write a minimizer f^* and residual $h_1 = \delta - d_0 f^*$.

- $d_0 f^*$ = removable inconsistency (gauge / exact component)
- h_1 = irreducible inconsistency (obstruction / residual)
- $\Phi_1 = \|h_1\|_1$ = total irreducible obstruction

4.3 Residual Quotient Obstruction (Non-closed Algebra)

The always-defined algebraic object mathematically modeling the irreducible residual is the **Residual Quotient space**: $Q^1(P) := C^1(P)/\,\mathrm{im}(d_0)$. The closure map $\kappa^1 : Q^1(P) \to C^2(P)$ given by $\kappa^1([h]) := d_1 h$ is well-defined. The condition $d_1 h_1 \neq 0$ measures the **curvature/closure charge** of the residual class.

4.3.1 The Core Sector Split

Inside $Q^1(P)$, the structure partitions:

- **closed sector (cohomology):** $ZQ^1(P) := \ker(\kappa^1) = H^1(P)$
- **open sector (topological faulting):** $\mathrm{im}(\kappa^1) = d_1(C^1(P)) =: \mathrm{Curv}^1(P) \subset C^2(P)$

This implies three core topological invariants:

- **quotient mass:** $\Phi_1([h]) = \inf_\alpha \|h + d_0\alpha\|_1$
- **closure charge:** $\Omega_2([h]) = d_1 h$
- **closure mass:** $\Gamma_2([h]) = \|d_1 h\|_1$

4.3.2 The Exact Sequence

This explicitly locks the global geometric behavior into the rigorous exact sequence:

$$0 \to H^1(P) \to Q^1(P) \xrightarrow{\kappa^1} \mathrm{im}(d_1) \to 0 \tag{4}$$

All evaluated h_1 components intrinsically define **quotient obstruction classes**. *Only* explicitly closed configurations formally represent **cohomology classes**.

4.4 The Two-Axis Obstruction Signature (Φ_1 and Γ_2)

We designate the pair (Φ_1, Γ_2) strictly as an independent **two-axis obstruction signature**.

Theorem 4.1 ($\Phi\Gamma$ Obstruction Signature Bounds)**.** *For every isolated residual quotient class $[h] \in Q^1(P)$, the resulting geometric signature $Sig([h]) := (\Phi_1([h]), \Gamma_2([h]))$ satisfies:*

1. $\Phi_1([h]) \geq 0$ *and* $\Gamma_2([h]) \geq 0$.
2. $\Phi_1([h]) = 0 \iff [h] = 0$ *perfectly exactly in* $Q^1(P)$.
3. $\Gamma_2([h]) = 0 \iff [h] \in H^1(P)$ *representing uncompromised globally closed cycles.*
4. $\Gamma_2([h]) \leq \|d_1\|_{(1\to 1)} \cdot \Phi_1([h])$.
5. *Formally restricted to* $Q^1(P)/H^1(P)$, Γ_2 *guarantees mathematical norm-equivalence.*

4.5 Closure Ratio Invariant

For nonzero $[h] \in Q^1(P)$, define the *Closure Ratio*:

$$R_{cl}([h]) := \frac{\Gamma_2([h])}{\Phi_1([h])} = \frac{\|d_1 h\|_1}{\inf_\alpha \|h + d_0\alpha\|_1} \tag{5}$$

Theorem 4.2 (Closure Ratio Bound)**.** $0 \leq R_{cl}([h]) \leq \|d_1\|_{(1\to 1)}$

4.5.1 R_{cl}-Controlled Dynamics (Guardrail Flow)

Define R_{cl}**-controlled dynamics** as the rule that obstruction descent must not trade global repair for uncontrolled closure rupture. Accept transition T iff: $\Phi' < \Phi$ and $R' \leq R_{\max}$.

Status: DERIVED

The closure ratio R_{cl} supplies a dynamical guardrail on obstruction descent. Convergent repair sequences driving Φ_1 to zero necessarily drive closure failure to zero as well. Thus R_{cl} turns the $\Phi\Gamma$ signature from a static diagnostic into a stability law.

5 §1.4 — Disjoint Observer Consistency Proof (Resolving L1)

We transition to **Conditional Rigidity** by deriving the aggregation rule structurally. Aczél-type representation theorems reduce admissible aggregation laws to $J_E(x) = g^{-1}(\sum_e g(j_e(x_e)))$. Replica Extensivity plus standard hypotheses forces g to be linear. **Combined with edge decomposability**, this lifts to:

$$J_E(x) = \sum_e j_e(x_e). \tag{6}$$

5.1 The Observer Triple

An observer is a triple: $\mathcal{O} := (\mathcal{A}, \mathcal{M}, \mathcal{J})$ where:

- $\mathcal{A}$ = aggregation law mapping multisets of local defect magnitudes to $\mathbb{R}_{\geq 0}$.
- $\mathcal{M} = \{M_e \subset \mathrm{End}(D_e)\}_e$ = admissible transformation monoid family.
- $\mathcal{J} = \{j_e : D_e \to \mathbb{R}_{\geq 0}\}_e$ = compatible local gauge family.

Theorem 5.1 (Conditional ℓ^1 Aggregation from Replica Additivity)**.** *RE selects ℓ^1 only under explicit structural assumptions: edge decomposability, permutation invariance, positive homogeneity, and convexity. Without these, non-linear Orlicz or ℓ^p-like aggregators survive.*

6 §1.5 — M-Compatible Obstruction Theory Classification

Foundationally reframed to define **M-compatible obstruction theory** as the parent geometrical framework, with ℓ^1 as the terminal/maximal-symmetry case.

Table 1: M-Compatible Classification Bounds

Allowed monoid (M_e)	Local diagnostic (j_e)	Global theory
All bounded linear maps	scalar multiple of Banach norm	weighted ℓ^1
All isometries	any isometry-invariant norm	symmetry-invariant additive
Graph automorphisms only	edge-orbit weights	orbit-weighted ℓ^1
Diagonal contractions	unconditional gauges	weighted ℓ^p / Orlicz

Theorem 6.1 (Maximal Symmetry Rigidity)**.** *Let $M = \mathrm{End}_b(V)$. If j is a faithful, homogeneous, subadditive, and M-compatible gauge, then $j(x) = c\|x\|$ for some $c > 0$.*

6.1 Formal Non-ℓ^1 Physical Models

6.1.1 Model I: Vector-Valued Bond Defects ($J_{2,1}$)

Local defect gauge is Euclidean norm $j_e(\omega_e) = c_e\|\omega_e\|_2$.

$$\Phi_{2,1}([\omega]) = \min_{\alpha} \sum_e \|\omega_e - (d_0\alpha)_e\|_2. \tag{7}$$

Testable Discrepancy: In a 3-node loop with circulation $c = (1, 1)$, scalar ℓ^1 predicts $\Phi = 2$, while $J_{2,1}$ predicts $\Phi = \sqrt{2}$.

6.1.2 Model II: Quantum Trace-Norm Inconsistency

Use trace distance $j_e(\delta_e) := \frac{1}{2}\|\delta_e\|_1$. The global quantum obstruction is $J_Q(\delta) := \sum_e \frac{1}{2}\|\delta_e\|_1$.

Theorem 6.2 (Quantum Trace-Norm Obstruction Duality)**.** $\Phi_Q([\delta]) = \max_\varphi \sum_e \mathrm{Tr}(\varphi_e\delta_e)$ *subject to $d_0^*\varphi = 0$, $\varphi_e = \varphi_e^\dagger$, and $\|\varphi_e\|_\infty \le 1/2$.*

7 §1.6 — Dual Observability and ℓ^1–ℓ^∞ Correspondence

Theorem 7.1 (Dual Observability)**.** *Let $\delta \in C^1(P)$ be the observed inconsistency. Then:*

$$\Phi([\delta]) = \min_{f \in C^0} \|\delta - d_0 f\|_1 = \max_{\varphi \in C^1} \{\langle \varphi, \delta\rangle : \|\varphi\|_\infty \le 1, d_0^*\varphi = 0\} \tag{8}$$

primal: minimize over gauge (remove what can be fixed)
dual: maximize observable inconsistency (detect what cannot)

8 §1.7 — Observer Passivity and Epistemic Quotients

An observer is a finite, deterministic, repair-compatible quotient map $\mathcal{O} : Q^1 \to Q^1_{\mathcal{O}}$.

Theorem 8.1 (Observer Passivity / No-Creation)**.** *Observers can erase or identify obstruction, but they cannot create it: $\Phi_{\mathcal{O}}(\mathcal{O}(q)) \le \Phi(q)$.*

Theorem 8.2 (No Omniscient Observer)**.** *No external observer can possess higher certified resolution than the core Q^1.*

9 §1.8 — Contextual Congruence and Witness-Rank

Definition 9.1 (Contextual Congruence)**.** $x \sim y \iff p(x) = p(y)$. This is a contextual congruence if $x \sim y \implies \rho(c)x \sim \rho(c)y$ for all contexts c. A **critical context witness** is $w = (x, y, c)$ where $x \sim y$ but $\rho(c)x \not\sim \rho(c)y$.

$$\boxed{\text{Obstruction alone does not force lift; context-visible non-congruence does.}}$$

10 §1.9 — Reflective Completion (Conditional)

Completion is a conditional reflection. It exists only under **Admissibility Hypotheses H1–H5** (Finite witness enumeration, finite search class, termination, minimality, preservation).

Theorem 10.1 (Conditional Reflection)**.** *Assume H1–H5. Then the inclusion* $I : \mathsf{Cong}^{adm} \hookrightarrow \mathsf{Raw}^{adm}$ *admits a left adjoint* $F^{ref} \dashv I$.

11 §1.10 — Licensed Lift Contracts

A valid lift contract Λ satisfies witness validity, lift morphism validity, and context compatibility.

Theorem 11.1 (Lift Contract Soundness)**.** *A valid lift contract cannot silently leave the original witnessed non-congruence unresolved and uncertified.*

12 §1.11 — Summary: What Is Derived and What Is Injected

Table 2: DERIVED WITHIN STATED ASSUMPTIONS

Structure	Source
Evaluation Preorder P	Reflexive-transitive closure of comparison
Cochain complex	Simplicial structure + coboundary
Dual representation Φ	LP strong duality
Observer passivity	Induced quotient norm
Witness-rank invariance	Linear independence in Q^1
Reflective completion	H1–H5 + initiality of lifts

Table 3: INJECTED (explicit assumptions)

Structure	When Injected
Scalar carrier ($\mathbb{R}$)	§1.0 — for specific $D = y - x$
Edge-decomposable aggregation	§1.4 — for ℓ^1 over edges
Admissibility hypotheses H1–H5	§1.9 — for reflective completion

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